

PowerBI - BankingAB Project

Core Banking System (CBS) Analytics — Power BI Portfolio Project

Developed by: Nishkarsha Khot | Oracle PL/SQL Developer | Power BI Developer | BFSI Domain | 3+ Years Experience

Tools: Power BI Desktop, Power BI Service, Oracle SQL, DAX, Data Gateway

Project Overview

An end-to-end Power BI portfolio project built on Core Banking System (CBS) data. Demonstrates data modeling, DAX, and multi-page dashboard development for the BFSI domain.

Attribute	Details
Domain	BFSI - Core Banking System (CBS)
Source	Oracle SQL Server (CBS e-Banker, 10 tables)
Target Audience	Bank Managers, Branch Officers, Auditors, MIS Teams
Deliverables	2 Power BI Reports, 5 Report Pages

Source Data (10 Tables)

Transactions, Loan Accounts, Deposit Accounts, Members, NPA Classification, GL Posting, Branch, Date Table, Product Types, EOD Batch Log.

Key Tasks

1. Connect & Load Data

- Connect to Oracle SQL Server (CBS e-Banker); load all 10 tables into Power Query.
- Fix data types (TxnDate, DisbursementDate, MaturityDate → Date).
- Create a StartDate parameter for incremental refresh; clean nulls/duplicates.

2. Data Modeling & Relationships

- Build star schema: Date, Members, Branch, Product Types as dimensions.
- Relate Transactions ↔ Date (TxnDateKey), Members (MemberID), Branch (BranchCode).
- Use USERELATIONSHIP for DisbursementDate vs TxnDate analysis; validate no circular joins.

3. Hierarchies & Fiscal Year

- Date Hierarchy: FY Year → FY Quarter → Month → Week → Weekday.
- Product Hierarchy: Category → Type → Name. Geography Hierarchy: Zone → State → Branch → City.
- Derive FY-Year and FY-Quarter with DAX (Indian fiscal year, April–March); mark Date Table as official.

4. Calculated Columns & Core DAX Measures

- Interest Income, NPA Status Label, Overdue Amount, Transaction Category (Credit/Debit).
- Core measures: Total Loan Disbursed, Total Deposits, Total Interest Income, NPA Ratio %, Credit-Deposit Ratio.

5. Data Validation

- Reconcile transaction counts and loan disbursements against source CBS reports.

- Verify GL double-entry balance (Total Debits = Total Credits).

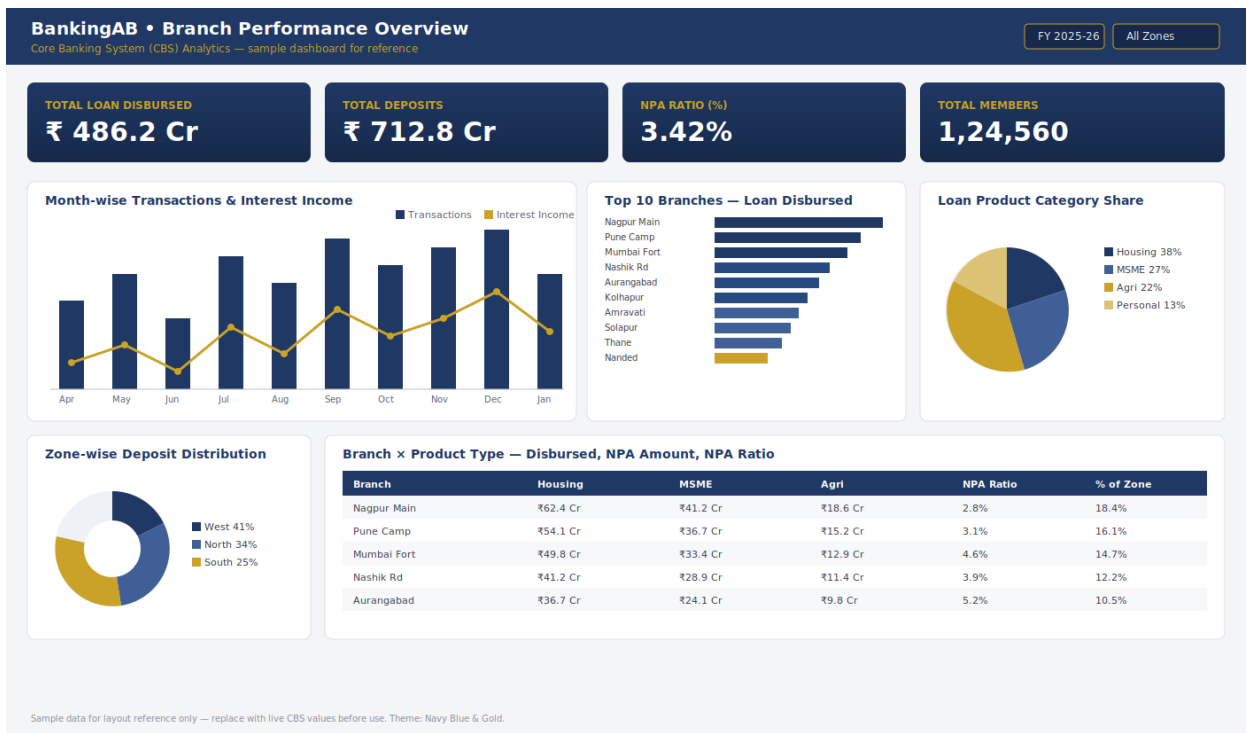
6. Dashboard Pages

- Branch Performance Overview — KPI cards, combo chart, top-10 branches, product & zone charts, drill-through.
- Branch Wise Analysis — loan/deposit/NPA trends, waterfall chart, state heat map.
- NPA & Risk Analysis — NPA classification, risk trend, branch scatter, DPD funnel chart.
- Profitability & Income Analysis — interest income trends, product/zone comparisons, Q&A visual.
- Custom tooltips, bookmarks (clear filters), and page navigation across all pages.

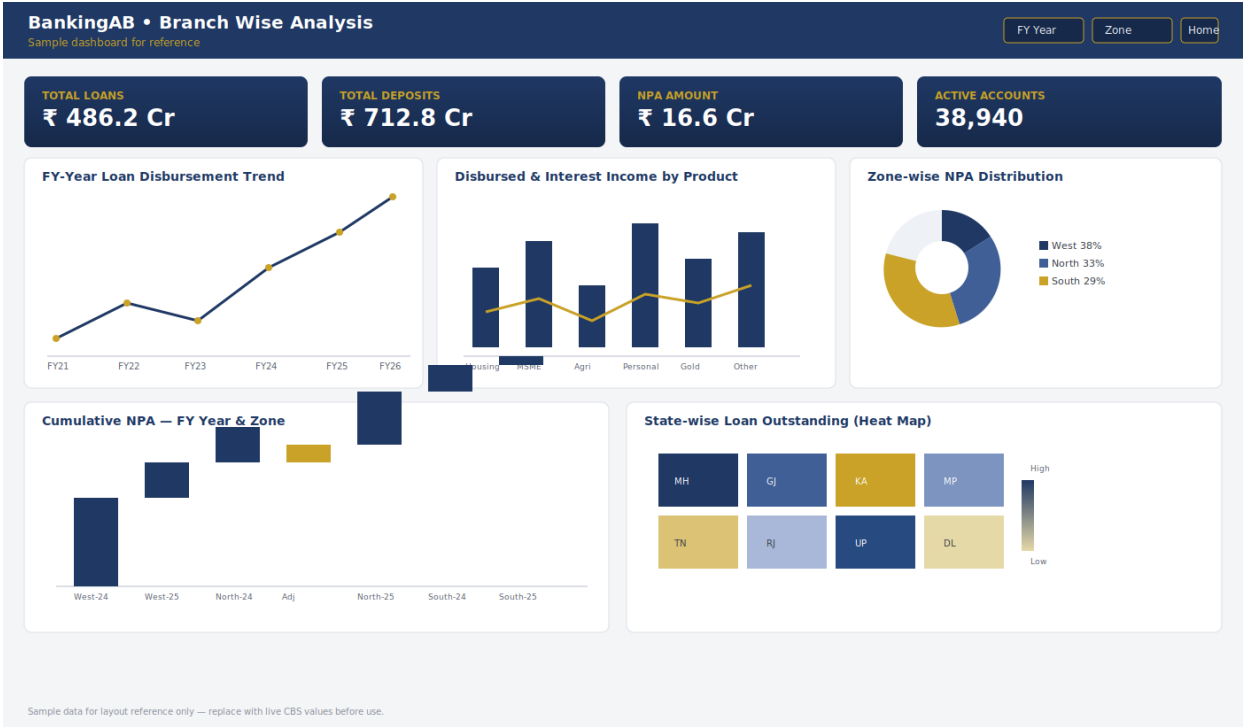
Sample Dashboards — Reference & Practice

Reference mockups for all 4 dashboard pages plus the custom tooltip. Use these as layout guides while building — replace sample numbers with live CBS values.

1. Branch Performance Overview (Main Dashboard)



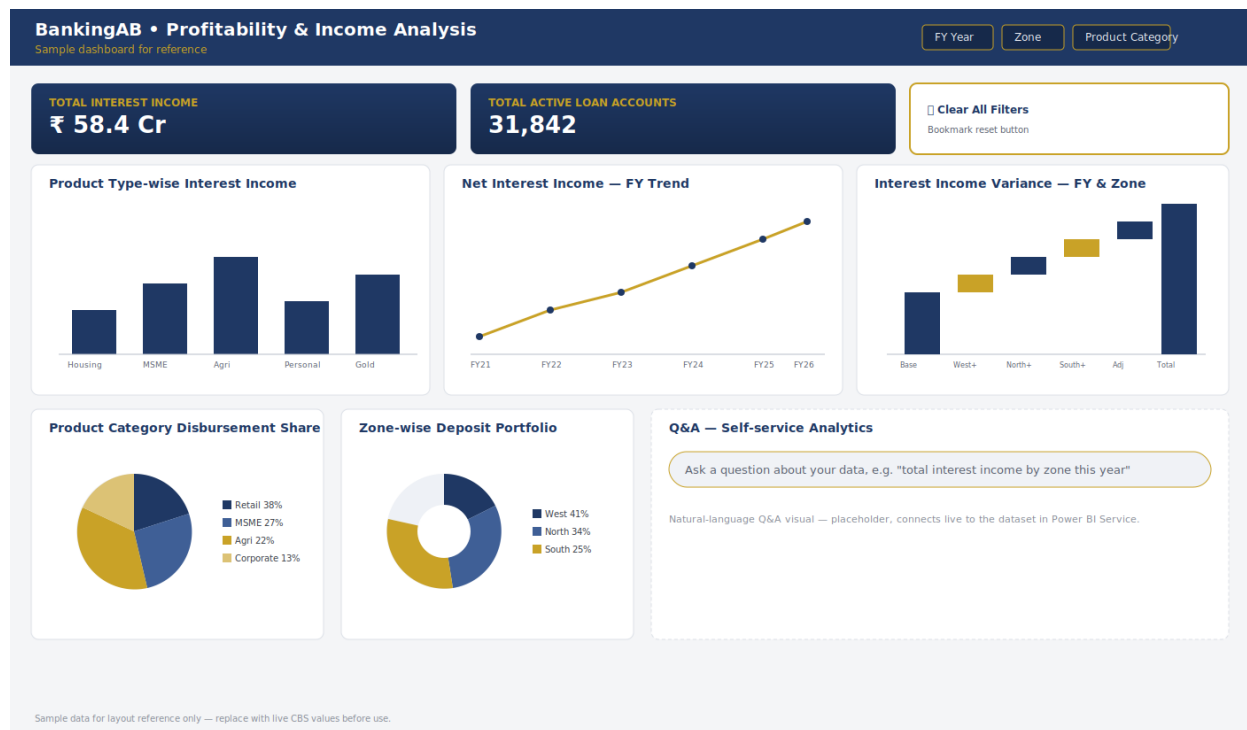
2. Branch Wise Analysis



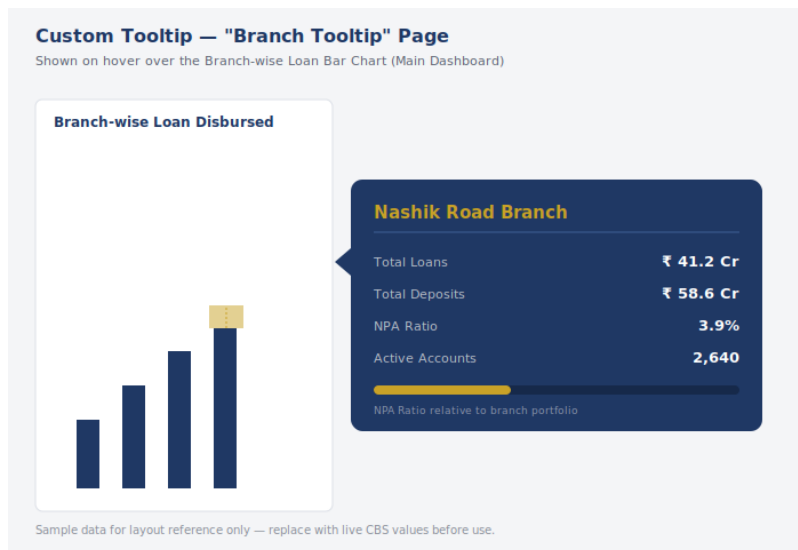
3. NPA & Risk Analysis



4. Profitability & Income Analysis



Branch Tooltip (Custom Tooltip Page)



7. Power BI Service Deployment

- Publish to BankingAB Workspace; configure On-Premises Data Gateway for Oracle source.
- Set StartDate parameter, schedule daily refresh after CBS EOD batch.
- Apply Row-Level Security (RLS): Branch Manager, Zone Manager, Auditor, Head Office roles.

Key DAX Measures

Measure	Formula
Total Loan Disbursed	SUM(LoanAccounts[DisbursedAmt])

Total Deposits	SUM(DepositAccounts[DepositAmt])
NPA Ratio %	DIVIDE([Total NPA Amount],[Total Loan Disbursed],0)*100
YoY Growth %	DIVIDE([CY Disbursed]-[PY Disbursed],[PY Disbursed],0)*100
QTD Disbursement	TOTALQTD([Total Loan Disbursed], DateTable[Date])

Skills Demonstrated

- Oracle SQL data connection & CBS schema understanding.
- Power Query transformations and parameter-based incremental load.
- Star-schema data modeling with active/inactive relationships.
- DAX: CALCULATE, USERRELATIONSHIP, ALLEXCEPT, RANKX, TOTALQTD, DIVIDE.
- Power BI Service: Workspace, Gateway, scheduled refresh, Row-Level Security.
- BFSI domain knowledge: NPA classification (RBI norms), Indian fiscal year, key banking KPIs.

Facilitator: Nishkarsha Khot | Oracle PL/SQL Developer | Power BI Developer | BFSI Domain Expert